## Missing Values

Charles River: Shipment table (Order-to-Cash)

| ShipmentNumber | ShipmentDate | FreightCost | TrackingNumber |
|---|---|---|---|
| SH-2024-000028 | 2024-01-15 | $189.01 | TRK202400000028 |
| SH-2024-000029 | 2024-01-16 | $35.31 | **(blank)** |
| SH-2024-000049 | 2024-01-21 | $106.73 | **(blank)** |

**Completeness problem:** TrackingNumber is blank, so the extract holds no carrier reference for these shipments.

## Duplicate Records

Charles River: PurchaseInvoice table (Procure-to-Pay)

| SupplierID | InvoiceNumber | InvoiceDate | GrandTotal |
|---|---|---|---|
| 2 | **V0002-2024-000182** | 2024-02-29 | $4,152.60 |
| 2 | **V0002-2024-000182** | 2024-03-04 | $3,454.20 |
| 2 | V0002-2024-000204 | 2024-03-17 | $950.16 |

**Potential duplicate:** The same supplier invoice number is recorded twice, with different dates and amounts. This is a duplicate-payment risk that requires investigation.

## Inconsistent Values

Charles River: Shipment table (Order-to-Cash)

| ShipmentNumber | ShipmentDate | Status | DeliveryDate |
|---|---|---|---|
| SH-2024-000001 | 2024-01-03 | Delivered | 2024-01-08 |
| SH-2024-000004 | 2024-01-06 | **In Transit** | 2024-01-10 |
| SH-2024-000038 | 2024-01-18 | **In Transit** | 2024-01-20 |

**Consistency problem:** These shipments are still marked In Transit, although their DeliveryDate passed years before the data was extracted.

## Outliers

Charles River: Shipment table (Order-to-Cash)

| ShipmentNumber | WarehouseID | FreightCost |
|---|---|---|
| SH-2024-000001 | 2 | $98.03 |
| SH-2024-000002 | 1 | $152.57 |
| SH-2024-000003 | 2 | $248.92 |
| SH-2026-008319 | 2 | **$873.95** |

**Outlier:** A FreightCost of $873.95 is several times the average shipment (about $150). It may be a legitimate large or rush shipment, so investigate before treating it as an error.